

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

UNIT I (BASICS OF OPERATIONAL AMPLIFIERS)

Degree, Semester & Branch: VI Semester B.E. ECE B

Course Code & Title: EC8453 & Linear Integrated Circuits

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: IC 741 Op-amp

Name of the Innovative Practice: Data Sheet of IC 741 Op-amp

Description:

An operational amplifier, also called as an op-amp or op amp, is an integrated circuit primarily designed for performing analogue computations. It has a very high voltage gain, typically of the order of 10^4 (100dB).

Although they are specially designed for performing operations like addition, subtraction, integration, differentiation etc., by using external components like resistors and capacitors to create a required feedback mechanism, it can also be used as an amplifier and for many other functions like filters, comparators etc.

Op-Amp IC's have become an integral part of almost all analogue circuitry. In this article we will take a look at one of the most used Op-Amp ICs: IC 741 Op Amp.

Many electrical parameters such as input offset voltage, input offset current, input bias current, maximum operating temperature etc. including the package details also mentioned in the data sheet. These data sheet are used in the IC Testing Industry to check the parameters after fabrication.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	-	-	-	-	-	1	1	1	-	-

CO1: The students will be able to design linear and non-linear applications of op-amps

References:

1. <http://www.ti.com/lit/ds/symlink/lm741.pdf>